CSE461

Introduction to Robotics Lab

Lab No. : 8
Group : 3
Section : 2
Semester : Fall_2025

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Submitted to -
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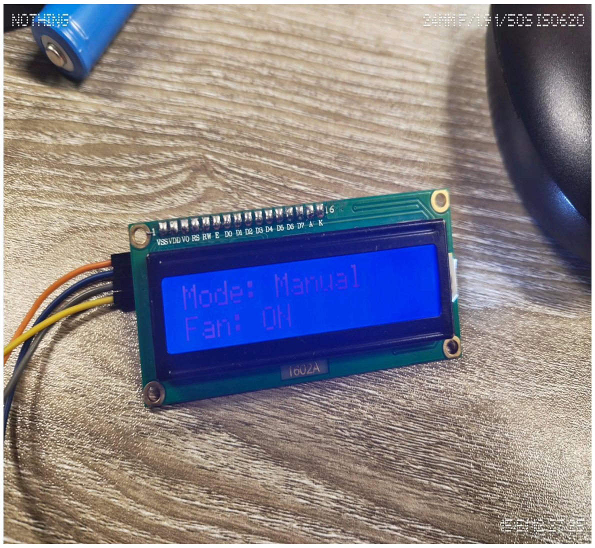
1. **Objectives:**
    - To design an environmentally friendly ventilation system that responds to light intensities autonomously.
    - Utilize pressing a button to switch between Automatic and Manual modes to add an alternative manual option.
    - To include several technological components into an operational model, like an LDR, DC motor, LED, and LCD.
    - To illustrate how one is capable of fixing programming logic and hardware connections.

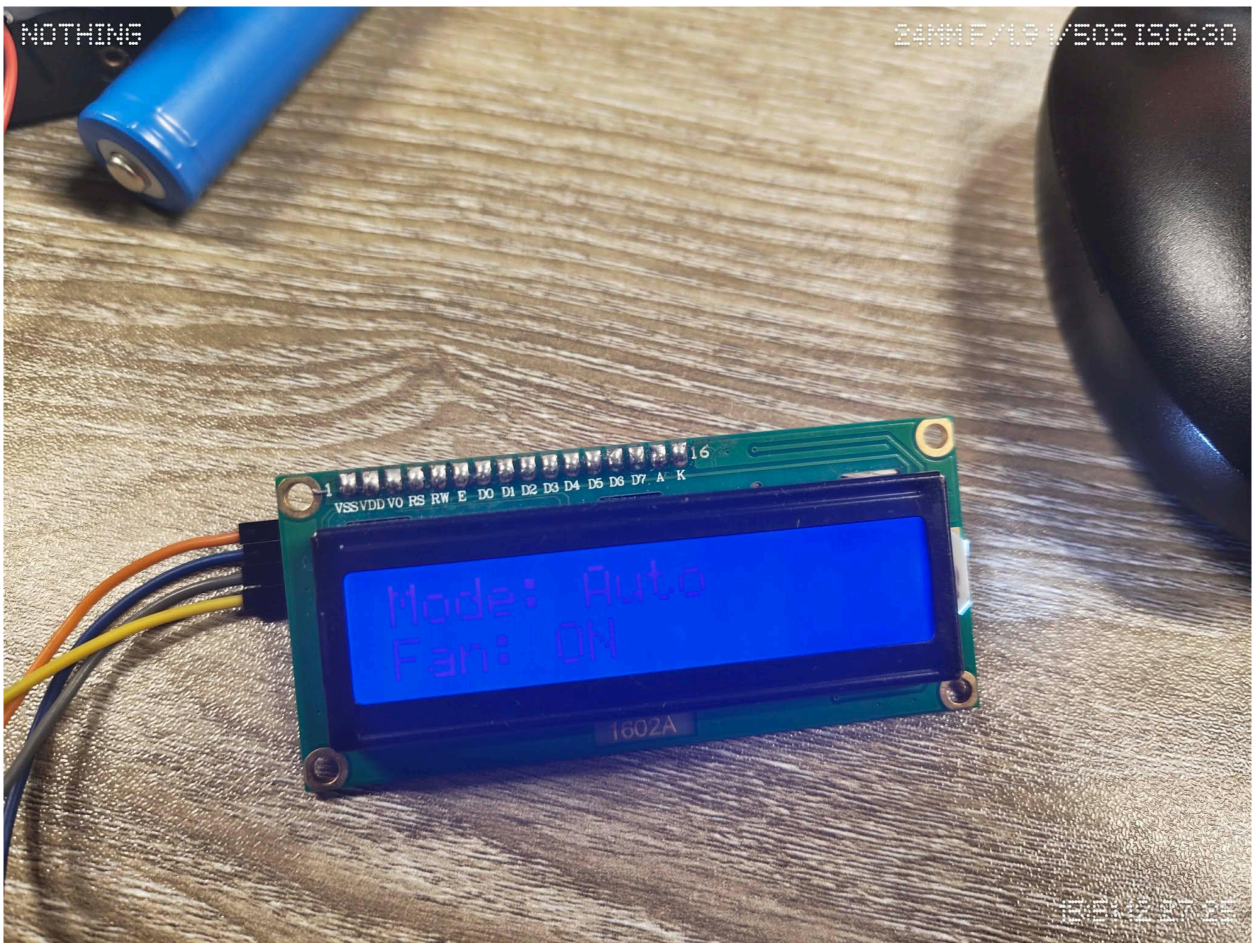
2. **Equipments:**
    - Arduino Uno
    - LDR
    - Push Button
    - DC Motor
    - LED
    - LCD Display with 12C module
    - Resistors
    - Connecting Wires and Breadboard

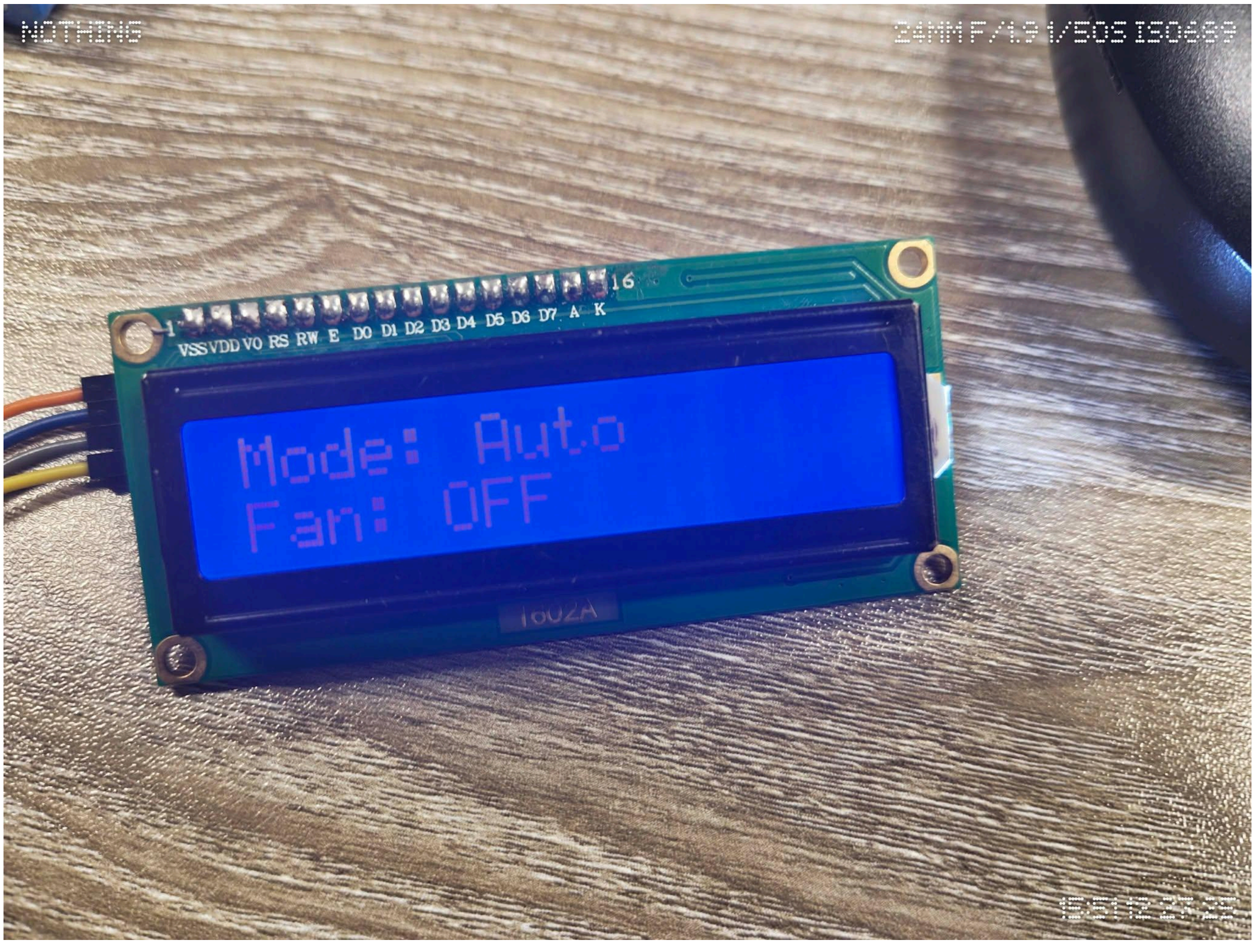
3. **Experimental Setup:**

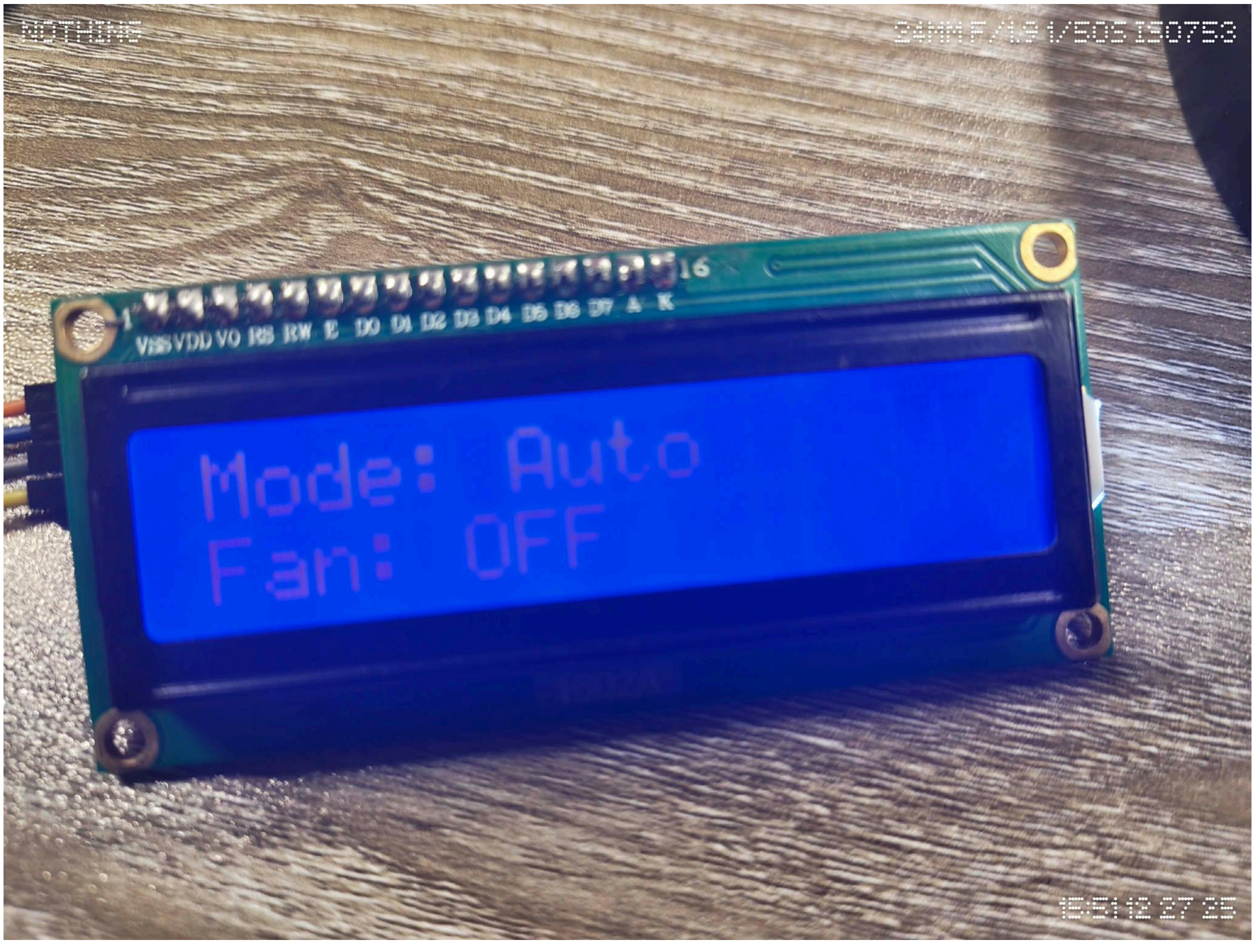
    (Picture + Explanation)

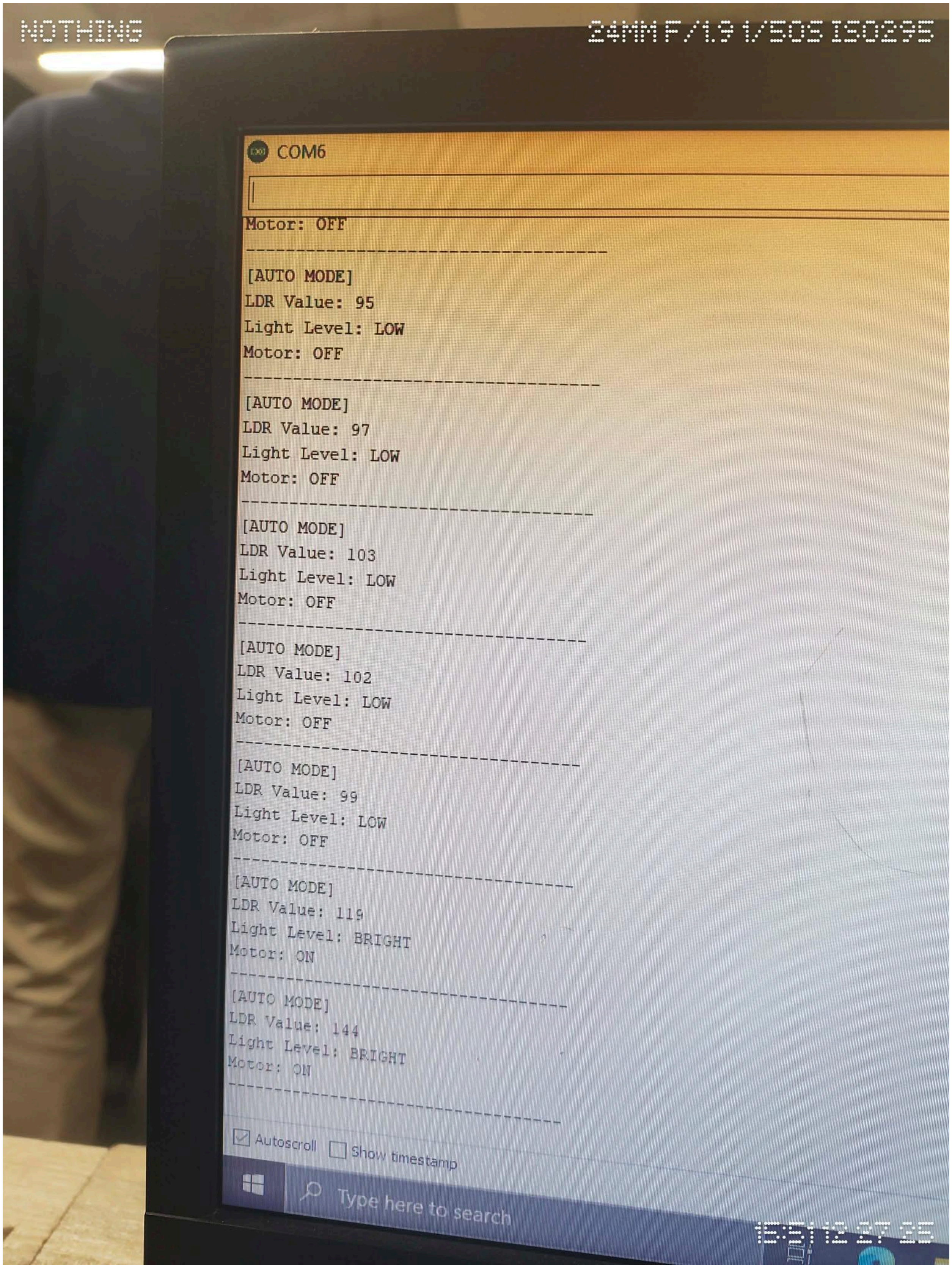
    There are two different modes of working included in the structure. Whenever the Arduino enters Auto Mode, it receives analog readings from the LDR and operates the DC motor and LED when the light intensity exceeds a set limit. The motor is switched on when using manual operation, regardless of the brightness of the light. In order to switch across these options, a push button works as a switching device. Continuous information on the motor's condition and the present setting can be seen on its LCD.


NOTHING
Mode: Manual
Fan: ON
1602A


NOTHING
24MM F/1.9 1/50S ISO630
1
16
VSS VDD V0 RS RW E D0 D1 D2 D3 D4 D5 D6 D7 A K
Mode: Auto
Fan: ON
1602A


NOTHING
VSS VDD V0 RS RW E D0 D1 D2 D3 D4 D5 D6 D7 A K
1
16
Mode: Auto
Fan: OFF
1602A


VSS VDD VO RS RW E D0 D1 D2 D3 D4 D5 D6 D7 A K
Mode: Auto
Fan: OFF


NOTHING
24MM F/1.9 1/50S ISO295
COM6
Motor: OFF

[AUTO MODE]
LDR Value: 95
Light Level: LOW
Motor: OFF

[AUTO MODE]
LDR Value: 97
Light Level: LOW
Motor: OFF

[AUTO MODE]
LDR Value: 103
Light Level: LOW
Motor: OFF

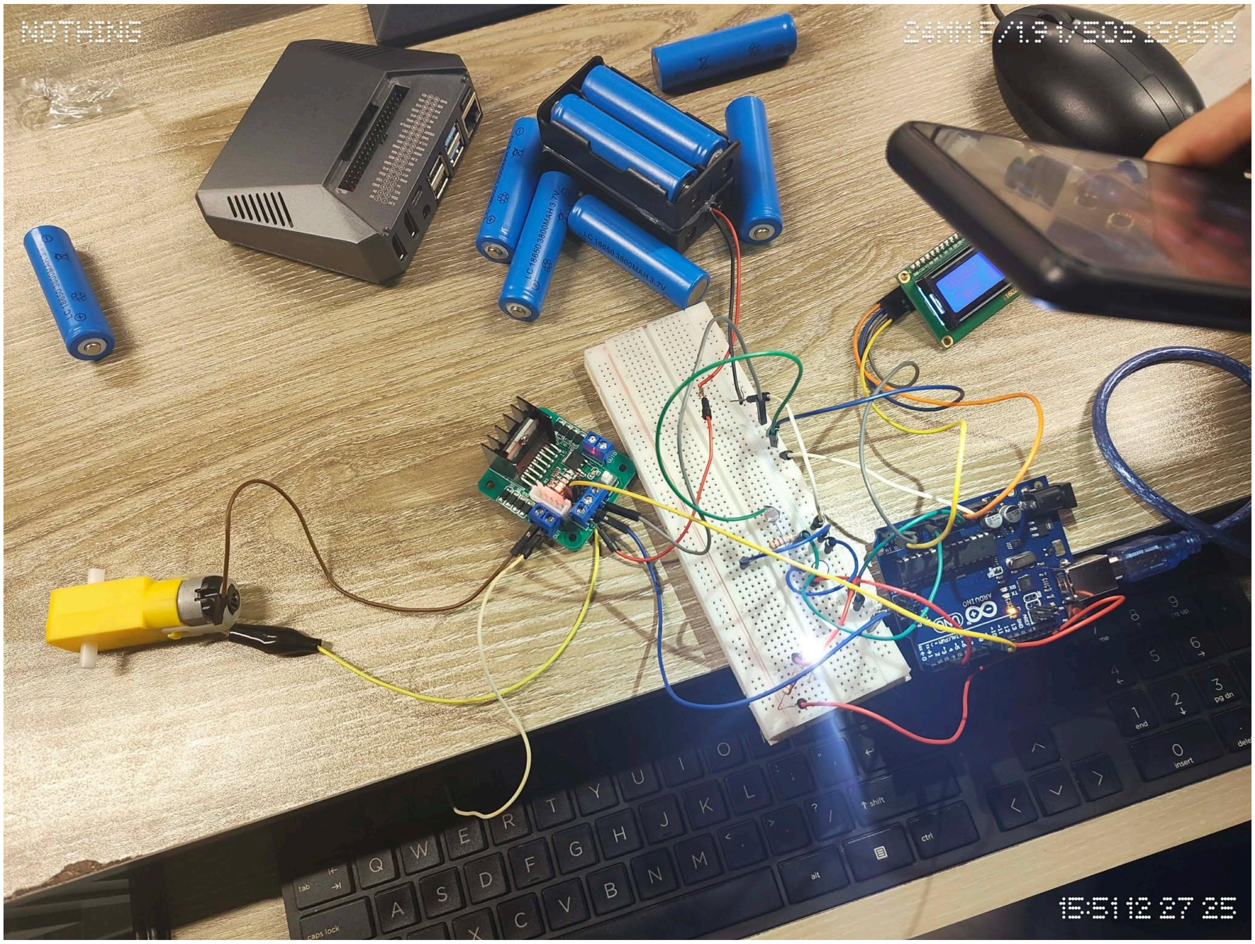
[AUTO MODE]
LDR Value: 102
Light Level: LOW
Motor: OFF

[AUTO MODE]
LDR Value: 99
Light Level: LOW
Motor: OFF

[AUTO MODE]
LDR Value: 119
Light Level: BRIGHT
Motor: ON

[AUTO MODE]
LDR Value: 144
Light Level: BRIGHT
Motor: ON

Autoscroll
Show timestamp
Type here to search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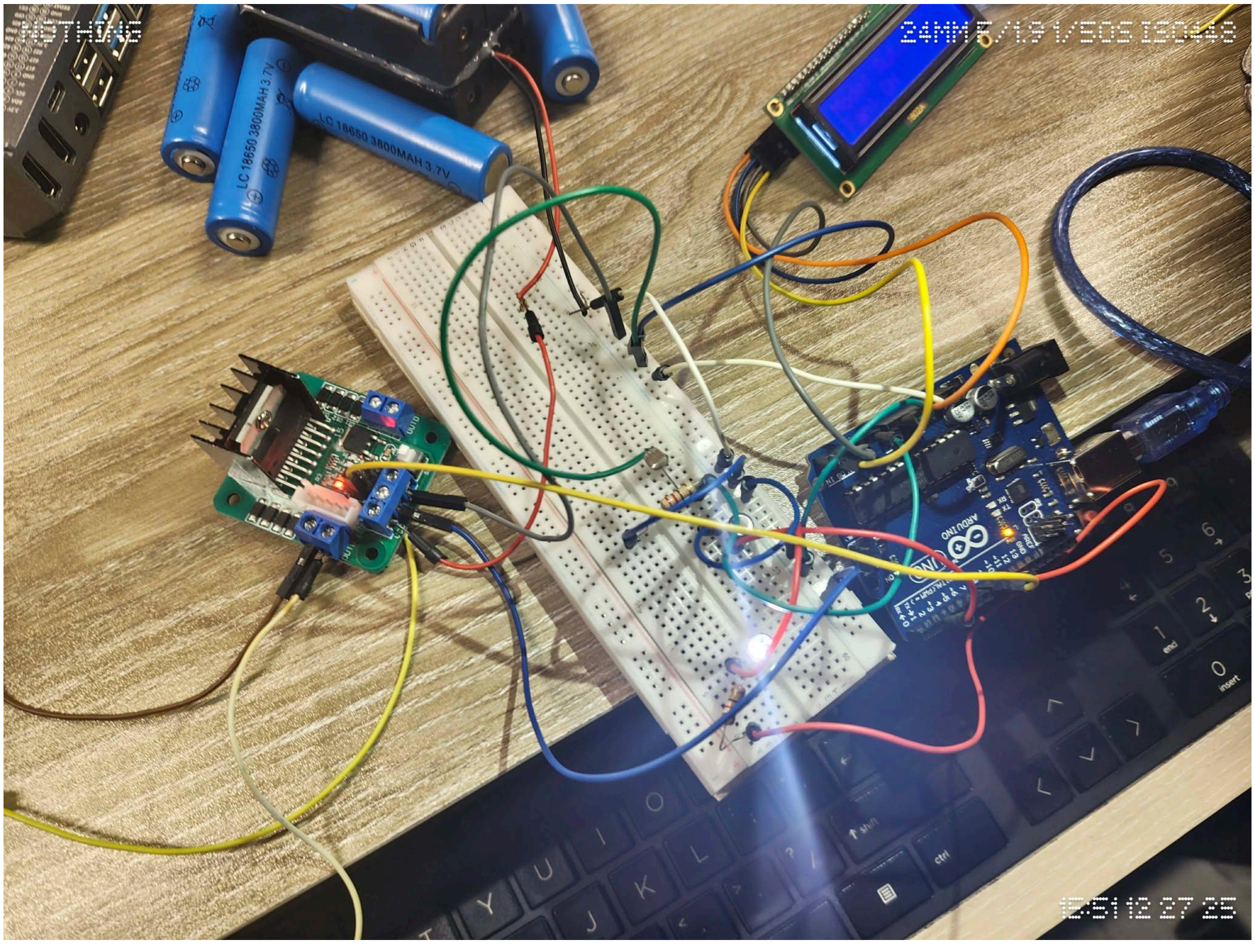
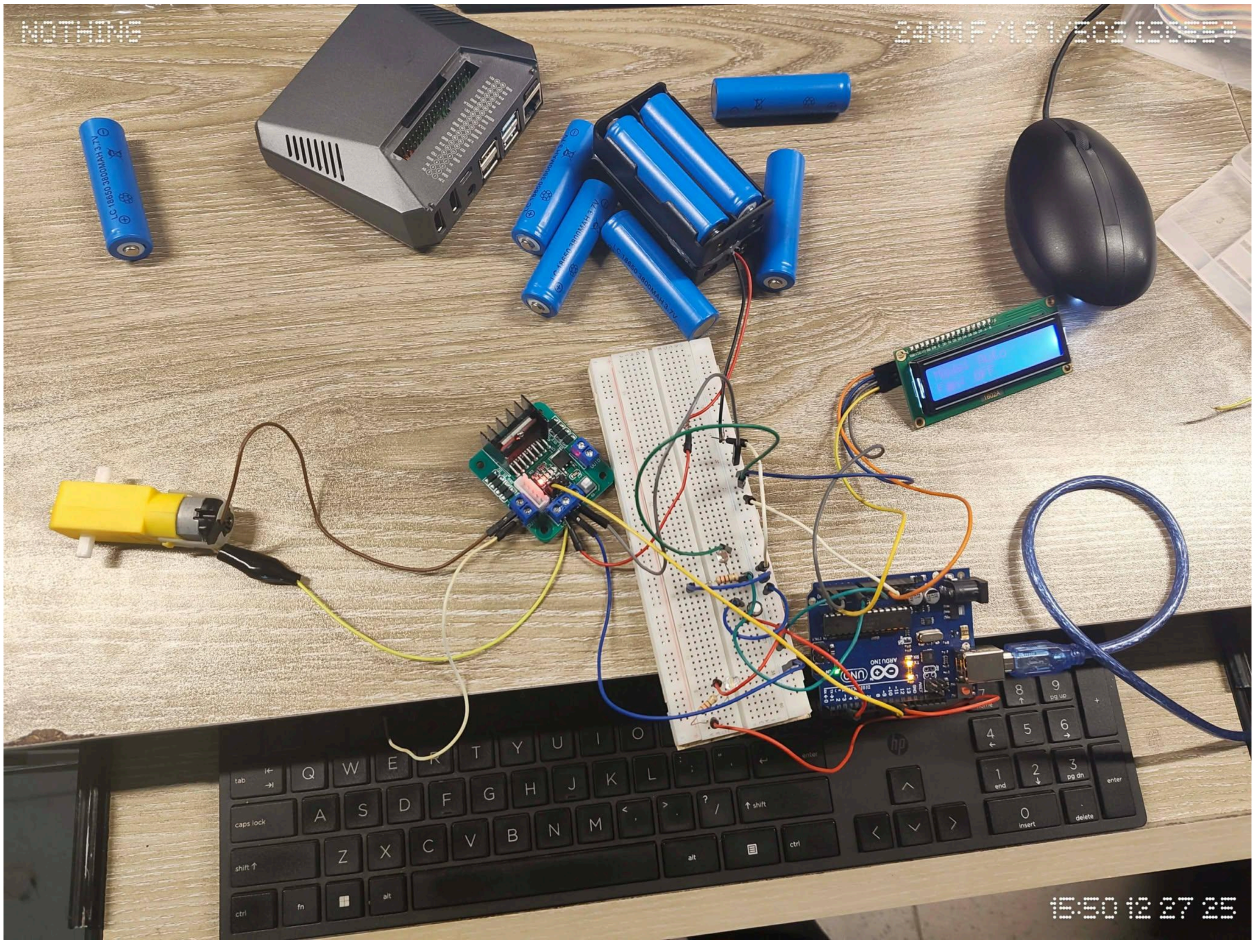
24MM F/1.9 1/50S ISO518
15:51 12 27 25


NOTHING
LC 18650 3800MAH 3.7V
LC 18650 3800MAH 3.7V
ARDUINO



4. Code: (If Applicable)

```
# Enter Code Here

#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// I2C LCD address (change to 0x3F if needed)
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pin definitions
const int ldrPin = A0;
const int buttonPin = 2;
const int motorPin = 8;
const int ledPin = 9;

// Variables
bool manualMode = false;
```

```
bool lastButtonState = HIGH;

int lightThreshold = 600;   // Adjust after seeing Serial Monitor readings

void setup() {
  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(motorPin, OUTPUT);
  pinMode(ledPin, OUTPUT);

  // Start Serial Monitor
  Serial.begin(9600);
  Serial.println("Environment-Aware Ventilation System");
  Serial.println("System Initializing...");
  Serial.println("----------------------------------");

  // Initialize LCD
  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Ventilation");
  lcd.setCursor(0, 1);
  lcd.print("System Ready");

  delay(2000);
  lcd.clear();
}

void loop() {

  // Read button
  bool buttonState = digitalRead(buttonPin);

  // Button press detection (toggle mode)
  if (lastButtonState == HIGH && buttonState == LOW) {
    manualMode = !manualMode;
    Serial.println("Button Pressed!");
    Serial.print("Mode Changed To: ");
    Serial.println(manualMode ? "MANUAL" : "AUTO");
    lcd.clear();
    delay(300); // Debounce
  }
  lastButtonState = buttonState;

  if (manualMode) {
    // MANUAL MODE
    digitalWrite(motorPin, HIGH);
    digitalWrite(ledPin, HIGH);
```

```
    lcd.setCursor(0, 0);
    lcd.print("Mode: Manual ");
    lcd.setCursor(0, 1);
    lcd.print("Fan: ON      ");

    Serial.println("[MANUAL MODE]");
    Serial.println("Motor: ON (LDR Ignored)");
  }
  else {
    // AUTO MODE
    int ldrValue = analogRead(ldrPin);

    lcd.setCursor(0, 0);
    lcd.print("Mode: Auto   ");

    Serial.println("[AUTO MODE]");
    Serial.print("LDR Value: ");
    Serial.println(ldrValue);

    if (ldrValue > lightThreshold) {
      digitalWrite(motorPin, HIGH);
      digitalWrite(ledPin, HIGH);
      lcd.setCursor(0, 1);
      lcd.print("Fan: ON      ");

      Serial.println("Light Level: BRIGHT");
      Serial.println("Motor: ON");
    }
    else {
      digitalWrite(motorPin, LOW);
      digitalWrite(ledPin, LOW);
      lcd.setCursor(0, 1);
      lcd.print("Fan: OFF     ");

      Serial.println("Light Level: LOW");
      Serial.println("Motor: OFF");
    }
  }

  Serial.println("----------------------------------");
  delay(1000);   // 1 second delay for readable Serial output
}
```

5. **Results (Output of the experiment):**

   The LED went on, and the DC motor started to rotate whenever the LDR was exposed to bright light. The motor and LED were turned off whenever they were concealed. The LCD switched to "Mode: Manual" after the button was pushed, and the fan proceeded to run repeatedly despite changes in lighting. Efficiently switched between "Mode: Auto" and "Mode: Manual" in accordance with feedback from the user.

6. **Discussions/Answers:**

   The demonstration effectively addressed a real-life issue by combining sensor data with actuator results. The primary problem was "switch bouncing," in which just one button push was recorded as many clicks. This issue was resolved by incorporating a 300-second resetting delay into the code. This study demonstrates the necessity of manual switches in computerized systems for ensuring accessibility to users in the case of sensor malfunction or certain environmental constraints.